

1 **AI-based prediction and monitoring system** 2 **for acute myocardial infarction detection** 3 **using laboratory biomarkers: a comparative** 4 **machine learning study with temporal** 5 **feature engineering**

6 **Hussam Aziz Ayed Alshammari¹ and Abdullah Eyada Mutlaq Alshammari¹**

7 ¹**Artificial Intelligence and Data Science Department, College of Computer Science and**
8 **Engineering, University of Hail, Hail 2440, Saudi Arabia**

9 Corresponding author:

10 Hussam Aziz Ayed Alshammari¹

11 Email address: Husam.a.ayed@gmail.com

12 **ABSTRACT**

13 Early and accurate detection of myocardial infarction (MI) is critical for timely clinical intervention and
14 improved patient outcomes. This study presents an AI-based prediction and monitoring system for acute
15 MI detection using laboratory biomarkers, specifically troponin and creatine kinase-MB (CK-MB). The
16 proposed system is designed to support clinical decision-making by providing automated risk stratification
17 and continuous patient monitoring across multiple hospital visits. A rigorous nested cross-validation
18 methodology with patient-level stratification is employed to ensure unbiased performance estimation
19 and prevent data leakage. This study compares four machine learning classifiers—Logistic Regression,
20 Random Forest, Histogram Gradient Boosting (HGB), and XGBoost—on a clinical dataset comprising
21 2,327 hospital visits from 1,506 unique patients. Temporal feature engineering techniques, including
22 delta changes between consecutive biomarker measurements, maximum prior values, and visit-level
23 tracking, enable continuous patient monitoring. The XGBoost classifier achieved the highest performance
24 with a precision-recall area under curve (PR-AUC) of 0.986 and a receiver operating characteristic area
25 under curve (ROC-AUC) of 0.982, while maintaining a clinically mandated target recall of 98%. Ensemble
26 methods (XGBoost, HGB, Random Forest) substantially outperformed Logistic Regression, achieving
27 F1-scores exceeding 0.94 compared to 0.87. The results demonstrate that modern gradient boosting
28 methods can serve as effective clinical decision support tools for MI detection, with particular strength in
29 exploiting temporal biomarker dynamics for continuous patient monitoring applications.

30 **Keywords:** Myocardial infarction; machine learning; XGBoost; cardiac biomarkers; clinical decision
31 support; temporal feature engineering; nested cross-validation; troponin; CK-MB; random forest

32 **INTRODUCTION**

33 Myocardial infarction (MI), commonly known as heart attack, remains one of the leading causes of
34 mortality worldwide (Thygesen et al., 2018). Early detection and timely intervention are crucial factors
35 in reducing MI-related morbidity and mortality. The diagnosis of MI traditionally relies on clinical
36 symptoms, electrocardiogram (ECG) findings, and elevated cardiac biomarkers, particularly cardiac
37 troponin (Amsterdam et al., 2014; Collet et al., 2021).

38 Cardiac troponin has become the gold standard biomarker for MI diagnosis due to its high sensitivity
39 and specificity for myocardial injury (Reichlin et al., 2009; Chapman et al., 2020). Creatine kinase-MB
40 (CK-MB), while less specific than troponin, continues to serve as a valuable complementary marker in
41 clinical practice (Aydin et al., 2019). The interpretation of these biomarkers, however, can be challenging
42 due to their time-dependent release kinetics and the need to consider serial measurements.

In emergency department settings, clinicians face the critical challenge of making rapid diagnostic decisions under time pressure and high patient volumes. The accurate interpretation of laboratory biomarker patterns, particularly when monitoring patients across multiple visits, requires integration of complex temporal information that can exceed human cognitive capacity. Artificial intelligence (AI) systems offer a promising solution to augment clinical decision-making by automatically analyzing biomarker trajectories, identifying high-risk patterns, and providing real-time risk stratification to support physician judgment.

Machine learning (ML) approaches have shown promise in various medical diagnostic applications, including cardiovascular disease prediction (Attia et al., 2019; Rajkomar et al., 2019; Topol, 2019). These methods can capture complex non-linear relationships between multiple variables and potentially improve diagnostic accuracy beyond traditional rule-based approaches. When integrated into clinical workflows, AI-based decision support tools can reduce diagnostic delays, minimize missed diagnoses, and enable more efficient allocation of healthcare resources (Wiens et al., 2019).

Despite the growing interest in ML-based MI detection, several methodological challenges remain. Many existing studies suffer from data leakage due to improper cross-validation strategies that do not account for patient-level dependencies (Varoquaux and Cheplygina, 2022). Additionally, comparative studies of multiple ML algorithms on the same dataset with consistent evaluation methodology are relatively scarce.

This study addresses these gaps by presenting a rigorous comparative analysis of four ML algorithms—Logistic Regression, Random Forest, HGB, and XGBoost—for MI detection using cardiac biomarkers. A nested cross-validation framework with patient-level stratification is employed to ensure unbiased model selection and performance estimation. The primary objective is to identify the most effective ML approach for MI detection while maintaining a clinically relevant target recall of 98%.

The main contributions of this work are:

1. A comprehensive comparison of four ML algorithms for MI detection using a rigorous nested cross-validation methodology.
2. Implementation of patient-level stratification to prevent data leakage and ensure generalizable performance estimates.
3. Development of temporal feature engineering techniques that capture the dynamic nature of biomarker changes across multiple hospital visits.
4. Threshold optimization to achieve a target recall of 98%, suitable for clinical screening applications.
5. A clinical decision support framework designed to enhance healthcare delivery through automated risk stratification and continuous patient monitoring.

The proposed AI-based system has the potential to significantly enrich the healthcare sector by addressing several critical challenges: reducing diagnostic delays that can lead to increased mortality, providing 24/7 automated screening capability in understaffed facilities, enabling objective and consistent interpretation of complex biomarker patterns, and supporting evidence-based clinical decision-making in high-pressure emergency settings.

RELATED WORK

Machine learning applications in cardiovascular disease detection have grown substantially in recent years. Topol (2019) provided a comprehensive overview of how AI is transforming clinical medicine, while Attia et al. (2019) demonstrated the potential of deep learning for detecting various cardiac conditions from ECG signals, achieving remarkable accuracy in identifying previously undiagnosed conditions. Deep neural networks applied to ECG data have also shown promise for mortality prediction (Raghunath et al., 2020).

In the context of biomarker-based MI detection, several studies have explored ML approaches. Than et al. (2019) developed decision support tools combining clinical features with troponin values to improve risk stratification in patients with suspected acute coronary syndrome. Their work highlighted the importance of serial biomarker measurements in improving diagnostic accuracy. The kinetics of high-sensitivity cardiac troponin, including the characteristic rise-and-fall pattern, have been extensively studied as the cornerstone of MI diagnosis under the Fourth Universal Definition (Chapman et al., 2020; Collet et al., 2021). Body et al. (2021) demonstrated that undetectable troponin levels at presentation

could be used for rapid rule-out of MI using high-sensitivity assays, further establishing the value of biomarker-based decision pathways.

A recent systematic review by Alahmadi and Alshammari (2024) surveyed ML approaches for heart disease prediction, identifying troponin levels and ensemble methods as key components in achieving high diagnostic accuracy. The review emphasized the need for rigorous evaluation methodologies and external validation to ensure clinical applicability.

Ensemble methods have shown particular promise in medical diagnosis. Gradient boosting algorithms, including XGBoost (Chen and Guestrin, 2016) and LightGBM (Ke et al., 2017), have demonstrated superior performance in various healthcare prediction tasks. Random Forest (Breiman, 2001), one of the earliest ensemble learning methods, continues to serve as a robust baseline in clinical studies. These methods effectively handle heterogeneous data types and can capture complex feature interactions.

The importance of proper evaluation methodology in medical ML has been emphasized by several authors. Varoquaux and Cheplygina (2022) highlighted common pitfalls in ML studies, including inappropriate cross-validation strategies that lead to over-optimistic performance estimates. Nested cross-validation has been recommended as the gold standard for unbiased model selection and evaluation (Cawley and Talbot, 2010). The choice of evaluation metrics is equally critical: Saito and Rehmsmeier (2015) demonstrated that precision-recall curves are more informative than ROC curves for imbalanced datasets, while Hicks et al. (2022) provided comprehensive recommendations for evaluation metrics in medical AI applications. Furthermore, Obermeyer et al. (2019) highlighted the risk of algorithmic bias in healthcare applications, underscoring the importance of fairness-aware model development.

The responsible deployment of ML in clinical settings requires careful consideration of potential harms. Wiens et al. (2019) outlined a roadmap for responsible ML in healthcare, advocating for transparency, interpretability, and rigorous validation. Explainability methods such as SHAP values (Lundberg et al., 2020) have become essential tools for understanding model predictions and building clinician trust in AI-based decision support systems.

MATERIALS AND METHODS

Figure 1 presents an overview of the complete methodology employed in this study, from data collection through model evaluation.

Dataset Description

The study utilized a hospital diagnosis dataset containing 2,327 patient visits from 1,506 unique patients. Each record included clinical and laboratory measurements obtained during hospital visits. The dataset was de-identified by replacing patient identifiers with cryptographic hashes to protect patient privacy.

The target variable was defined as acute MI based on ICD-10 codes I21 (ST-elevation and non-ST-elevation myocardial infarction) and I22 (subsequent myocardial infarction), as well as keyword matching in diagnosis text including “myocardial infarction,” “STEMI,” and “NSTEMI.” Cases containing terms such as “history of” or “old MI” were excluded to focus on acute events. The dataset exhibited a positive rate of 57.5% at the visit level, with 647 positive patients and 859 negative patients.

Feature Engineering

The raw dataset was processed to extract clinically meaningful features. Numerical features included patient age, troponin levels, and CK-MB values. Categorical features included visit type. All numerical features underwent median imputation for missing values followed by standardization using z-score normalization.

To capture the temporal dynamics of biomarker changes, which are clinically important for MI diagnosis, several derived features were engineered:

- **Previous values:** The immediately preceding troponin and CK-MB measurements for each patient.
- **Delta changes:** The difference between current and previous biomarker values.
- **Maximum prior:** The maximum observed biomarker value up to the previous visit.
- **Visit index:** A sequential counter of visits for each patient.
- **Days since last visit:** The temporal gap between consecutive visits.

Table 1 presents the complete feature inventory used in this study.

This resulted in a total of 12 features: 11 numerical features and 1 categorical feature (visit type), which was one-hot encoded.

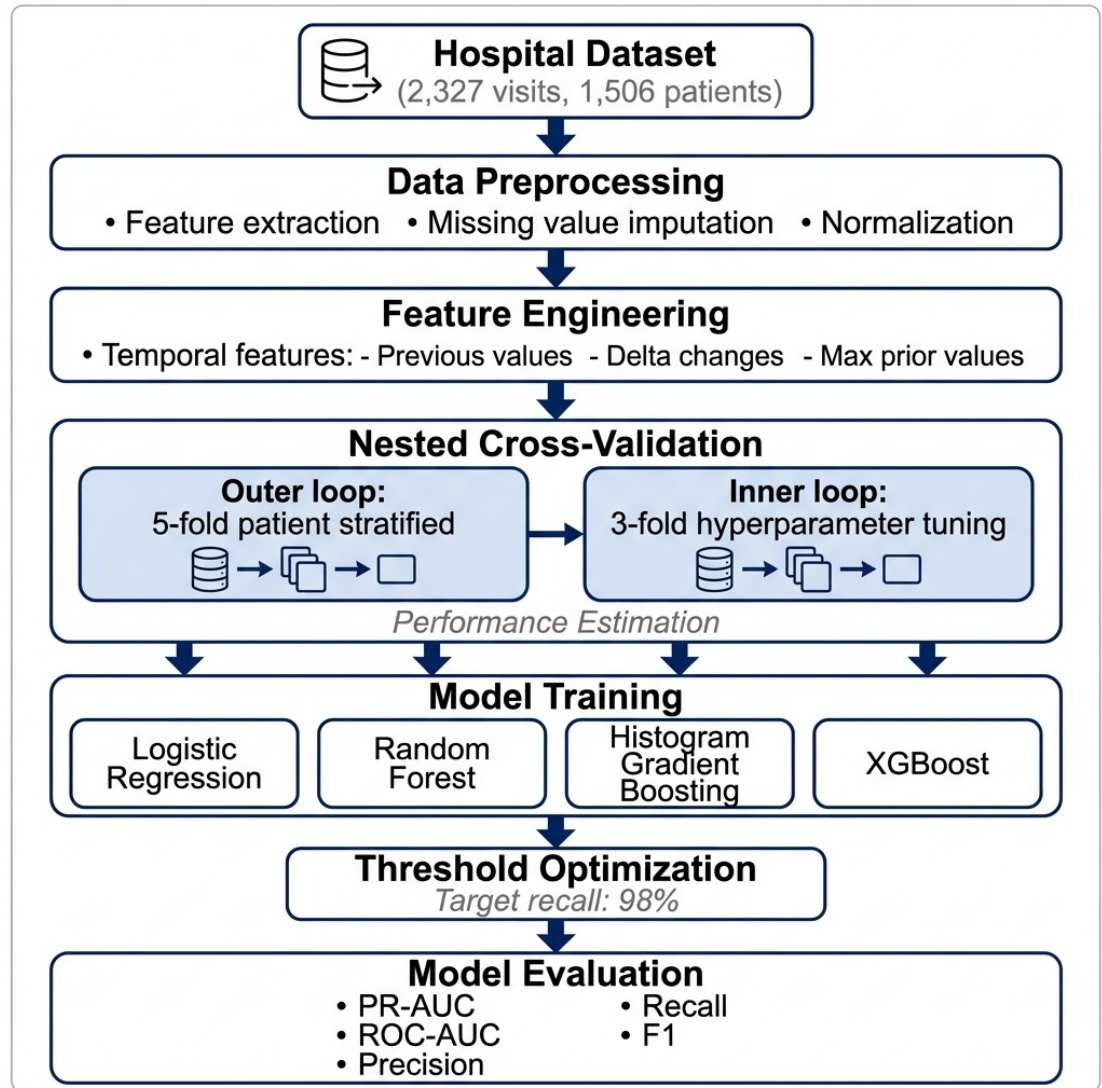


Figure 1. Workflow diagram of the proposed MI detection methodology. The pipeline includes data preprocessing, temporal feature engineering, nested cross-validation with patient-level stratification, training of four ML classifiers, threshold optimization for 98% recall, and comprehensive model evaluation.

Feature	Type	Description
age	Numerical	Patient age, standardized
visit_type	Categorical	Visit context (one-hot encoded)
visit_index	Numerical	Sequential visit counter per patient
days_since_last_visit	Numerical	Days between consecutive visits
troponin	Numerical	Current troponin level
troponin_prev	Numerical	Previous troponin measurement
troponin_delta	Numerical	Change from previous troponin
troponin_max_prior	Numerical	Maximum prior troponin value
ckmb	Numerical	Current CK-MB level
ckmb_prev	Numerical	Previous CK-MB measurement
ckmb_delta	Numerical	Change from previous CK-MB
ckmb_max_prior	Numerical	Maximum prior CK-MB value

Table 1. Complete feature inventory for the MI prediction model.

Machine Learning Models

Four ML classifiers were evaluated in this study:

Logistic Regression A linear classifier with L2 regularization. The regularization strength parameter C was tuned from 10^{-3} to 10^2 on a logarithmic scale. Class weights were balanced to account for class imbalance.

Random Forest An ensemble of 600 decision trees with bootstrap aggregation. Tuned hyperparameters included maximum tree depth (3, 5, 8, or unlimited), minimum samples per leaf (1, 5, 10, 30), and maximum features (square root, log2, or all). Balanced subsample class weighting was applied.

Histogram Gradient Boosting (HGB) A gradient boosting implementation optimized for large datasets using histogram-based splitting. Tuned hyperparameters included maximum depth (3, 4, 5, unlimited), learning rate (0.02–0.1), maximum leaf nodes (15–127), minimum samples per leaf (10–80), and L2 regularization (0, 0.1, 1.0).

XGBoost An optimized gradient boosting implementation with 900 estimators. Tuned hyperparameters included maximum depth (3–6), learning rate (0.02–0.08), subsample ratio (0.7–1.0), column subsample ratio (0.7–1.0), and minimum child weight (1, 3, 5).

Algorithm Formulations

Logistic Regression Logistic Regression models the probability of MI as a function of input features using the sigmoid function:

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (1)$$

where $\mathbf{w}$ represents the weight vector, $\mathbf{x}$ is the feature vector, and b is the bias term. With L2 regularization, the objective function becomes:

$$\min_{\mathbf{w}, b} \sum_{i=1}^n \mathcal{L}(y_i, \hat{y}_i) + \frac{1}{C} \|\mathbf{w}\|_2^2 \quad (2)$$

where C is the regularization strength parameter and $\mathcal{L}$ is the cross-entropy loss.

Random Forest Random Forest constructs an ensemble of B decision trees and combines their predictions through averaging:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B f_b(\mathbf{x}) \quad (3)$$

where $f_b(\mathbf{x})$ represents the prediction of the b -th tree. In this study, $B = 600$ trees were used with bootstrap aggregation to reduce variance and prevent overfitting.

172 **Gradient Boosting (HGB and XGBoost)** Gradient boosting methods build an additive model by
 173 iteratively fitting new trees to the residuals:

$$\hat{y}^{(t)} = \hat{y}^{(t-1)} + \eta \cdot f_t(\mathbf{x}) \quad (4)$$

174 where η is the learning rate and f_t is the tree added at iteration t . XGBoost minimizes the following
 175 regularized objective:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \quad (5)$$

176 where the regularization term is defined as:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (6)$$

177 with T being the number of leaves, w_j the leaf weights, and γ, λ the regularization parameters.

178 **Evaluation Metrics** Model performance was evaluated using the following metrics. Precision and recall
 179 are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

180 The F1-score provides the harmonic mean of precision and recall:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

181 The area under the precision-recall curve (PR-AUC) and receiver operating characteristic curve (ROC-
 182 AUC) provide threshold-independent performance measures:

$$\text{PR-AUC} = \int_0^1 P(R) dR \quad (9)$$

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}) \quad (10)$$

183 **Software Environment**

184 All experiments were implemented in Python 3.12.12, running on Google Colab. Key library versions
 185 were: scikit-learn 1.6.1, xgboost $\geq 1.7.0$ (installed via `pip install xgboost`), pandas $\geq 2.0.0$,
 186 numpy $\geq 1.24.0$, and matplotlib $\geq 3.7.0$. All stochastic operations were seeded with the fixed value `SEED`
 187 `= 42` to ensure full reproducibility.

188 **Evaluation Methodology**

189 A nested cross-validation framework was employed to obtain unbiased performance estimates while
 190 performing hyperparameter optimization (Cawley and Talbot, 2010). The outer loop consisted of 5-fold
 191 stratified cross-validation with patient-level grouping, ensuring that all visits from a given patient appeared
 192 in either the training or test set, but not both. The inner loop used 3-fold stratified cross-validation for
 193 hyperparameter tuning via randomized search with 12 iterations.

194 For each outer fold, the best hyperparameters were selected based on average precision score (equiva-
 195 lent to PR-AUC) in the inner validation. The final model was trained on the entire outer training set and
 196 evaluated on the held-out test set.

197 Decision thresholds were optimized to achieve a target recall of 98%, reflecting the clinical priority
 198 of minimizing missed MI diagnoses. For each fold, the threshold achieving at least 98% recall with
 199 the highest precision was selected from the precision-recall curve computed on inner fold validation
 200 predictions.

201 Performance metrics included PR-AUC, ROC-AUC, precision, recall, and F1-score, all computed at
 202 the optimized threshold.

Model	Hyperparameter Search Space
LogReg	C : 10^{-3} to 10^2 (12 log-spaced values)
RF	max_depth: {3, 5, 8, None}; min_samples_leaf: {1, 5, 10, 30}; max_features: {sqrt, log2, None}
HGB	max_depth: {3, 4, 5, None}; learning_rate: {0.02, 0.03, 0.05, 0.08, 0.1}; max_leaf_nodes: {15, 31, 63, 127}; min_samples_leaf: {10, 20, 40, 80}; L2_regularization: {0, 0.1, 1.0}
XGBoost	max_depth: {3, 4, 5, 6}; learning_rate: {0.02, 0.03, 0.05, 0.08}; subsample: {0.7, 0.85, 1.0}; colsample_bytree: {0.7, 0.85, 1.0}; min_child_weight: {1, 3, 5}

Table 2. Hyperparameter search spaces for each classifier.

Hyperparameter Search Space

Table 2 summarizes the hyperparameter search spaces explored during nested cross-validation. Model selection was performed using RandomizedSearchCV with 12 iterations per fold, optimizing for average precision score.

The XGBoost classifier was initialized with 900 estimators and the histogram tree method for computational efficiency. Random Forest used 600 estimators with balanced subsample class weighting. All random operations were seeded with the fixed value 42 ($SEED = 42$) to ensure reproducibility, consistent with the nested cross-validation structure (outer: 5-fold, inner: 3-fold) confirmed in the experimental implementation. The target recall of 98% was achieved by optimizing the classification threshold on the precision-recall curve from validation predictions.

Ethical Approval and Informed Consent

This study was approved by the Institutional Review Board (IRB) of the Hail Health Cluster, Research and Studies Department, Hail Region, Kingdom of Saudi Arabia (IRB Log Number: 2025-13; Category: Expedited; Date of Approval: February 24, 2025). The approval was granted for the study entitled “Cardiac Arrest Prediction and Monitoring Using Artificial Intelligence”. The study is retrospective in nature, utilizing a fully de-identified clinical dataset in which all direct and indirect patient identifiers were replaced with cryptographic hashes prior to any analysis. Informed consent from individual participants was not required by the IRB, given the retrospective and fully de-identified design of this study and in accordance with applicable data governance policy. A copy of the original signed IRB approval letter is provided as a confidential supplemental file accompanying this submission.

Use of Artificial Intelligence Tools

The authors used Claude (claude-sonnet-4-6, Anthropic, <https://claude.ai/>) for grammar correction and manuscript editing only. All figures in this manuscript were created programmatically using Python 3.12.12 with the matplotlib library (version $\geq 3.7.0$); no AI image-generation tools were used. No AI tool was used for data collection, statistical analysis, result generation, or scientific interpretation. All scientific content, methodology, experimental design, and conclusions are entirely the authors’ own work.

RESULTS

Cross-Validation Performance

Table 3 presents the nested cross-validation results for all four classifiers. XGBoost achieved the highest mean PR-AUC of 0.9857, followed closely by HGB (0.9851) and Random Forest (0.9815). Logistic Regression showed notably lower performance with a mean PR-AUC of 0.9646.

All models successfully achieved the target recall of approximately 98%. However, precision varied substantially across classifiers. The tree-based ensemble methods (Random Forest, HGB, XGBoost) achieved precision values exceeding 0.89, while Logistic Regression showed considerably lower precision (0.778) at the target recall level.

Figure 2 presents a visual comparison of all four classifiers in terms of their mean recall versus mean PR-AUC performance. The scatter plot clearly shows that gradient boosting methods (XGBoost and HGB) cluster in the upper region with the highest PR-AUC values, while Logistic Regression occupies the lower left quadrant.

Model	PR-AUC	ROC-AUC	Precision	Recall	F1
XGBoost	.986 $\pm$.005	.982 $\pm$.007	.898 $\pm$.021	.981 $\pm$.014	.937 $\pm$.010
HGB	.985 $\pm$.006	.981 $\pm$.008	.905 $\pm$.018	.979 $\pm$.015	.941 $\pm$.010
RF	.981 $\pm$.009	.980 $\pm$.008	.911 $\pm$.009	.982 $\pm$.014	.945 $\pm$.008
LogReg	.965 $\pm$.011	.961 $\pm$.012	.778 $\pm$.020	.981 $\pm$.013	.867 $\pm$.013

Table 3. Nested cross-validation performance comparison of ML classifiers (mean $\pm$ std over 5 outer folds).

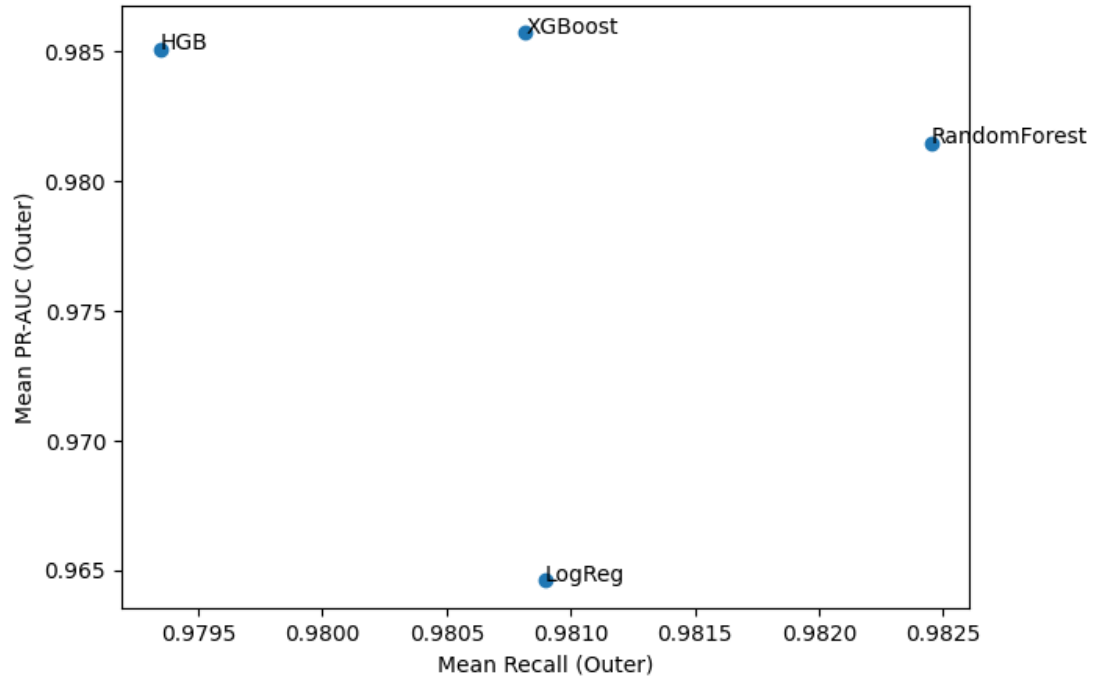


Figure 2. Model comparison showing mean PR-AUC versus mean recall for all four classifiers. XGBoost and HGB achieve the highest PR-AUC values while maintaining target recall levels.

Final Model Performance

Based on the nested cross-validation results, XGBoost was selected as the final model. The final XGBoost classifier, trained on the entire dataset with the most frequently selected hyperparameters, achieved:

- PR-AUC: 0.9845
- ROC-AUC: 0.9804
- Precision: 0.8968
- Recall: 0.9806
- F1-Score: 0.9368

The optimal operating threshold was determined to be 0.17. Table 4 presents the confusion matrix for the final model.

	Predicted Negative	Predicted Positive
True Negative	838	151
True Positive	26	1312

Table 4. Confusion matrix for the final XGBoost model at the optimized threshold of 0.17.

The model correctly identified 1,312 of 1,338 MI cases while missing only 26 cases, corresponding to

253 the target recall of 98.1%. Among the 989 non-MI cases, 838 were correctly classified and 151 were false
254 positives. Figure 3 provides a visual representation of these classification results as a heatmap.

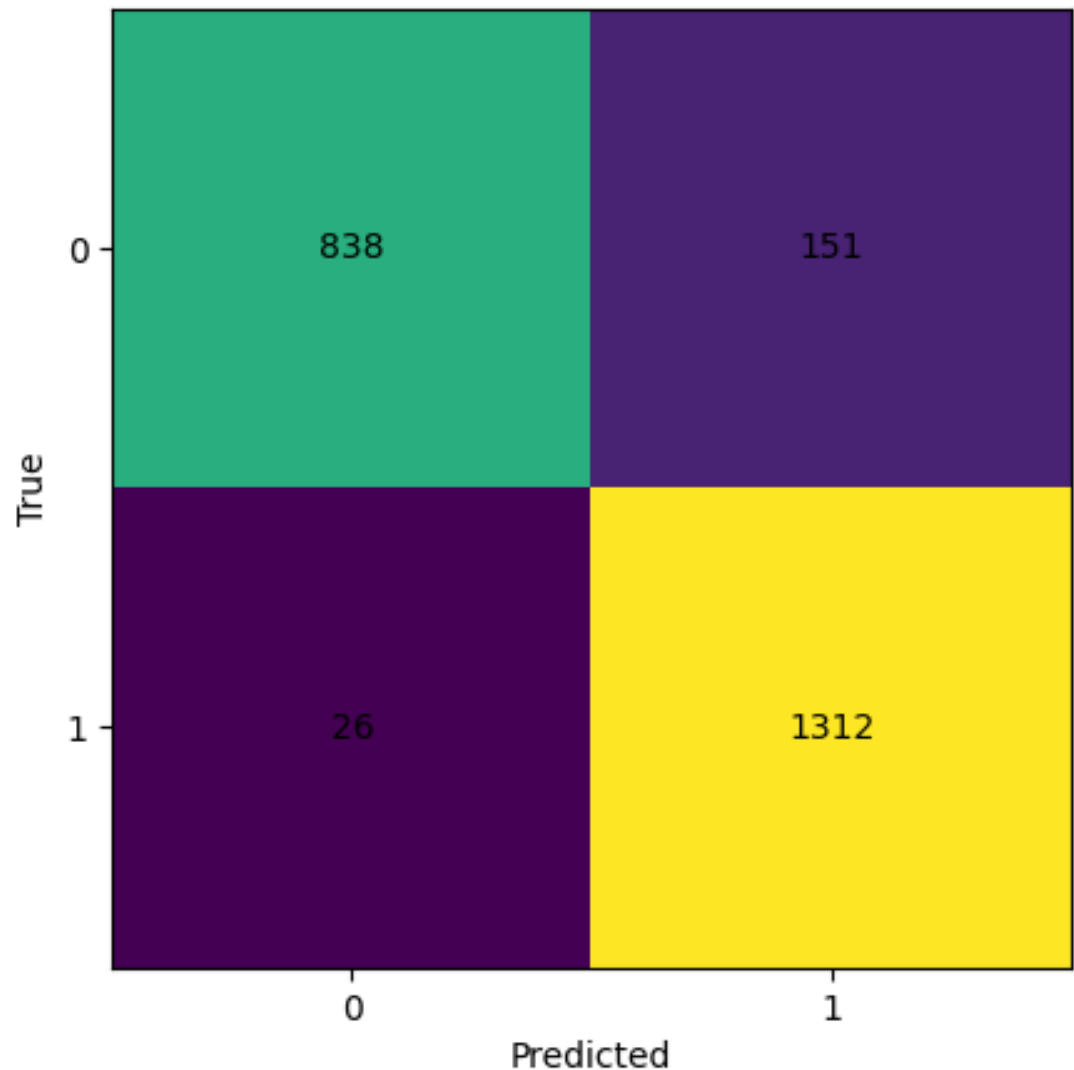


Figure 3. Confusion matrix heatmap for the final XGBoost model. The matrix shows strong diagonal dominance with 838 true negatives and 1,312 true positives, indicating excellent classification performance at the optimized threshold.

255 **Per-Fold Analysis**

256 Table 5 presents the per-fold results for XGBoost, demonstrating consistent performance across different
257 data splits. The decision threshold varied between 0.14 and 0.21 across folds, with all folds successfully
258 achieving the target recall.

259 **Performance Curves Analysis**

260 The precision-recall (PR) curve and receiver operating characteristic (ROC) curve provide comprehensive
261 insights into the model's discrimination ability across all possible threshold values.

262 Figure 4 presents the precision-recall curve for the final XGBoost model, demonstrating the trade-off
263 between precision and recall at different classification thresholds. The curve maintains high precision
264 (above 0.9) for recall values up to approximately 0.8, with a gradual decline thereafter as the model
265 captures more true positives at the cost of increasing false positives.

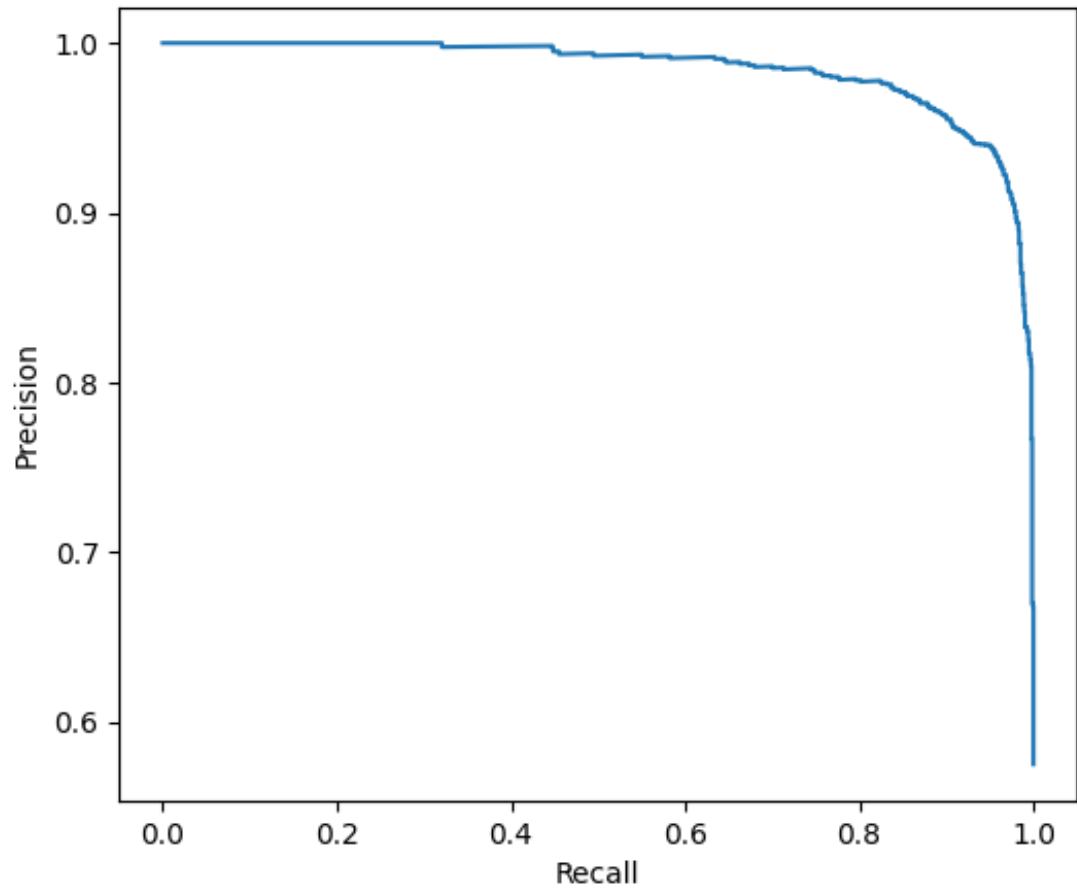


Figure 4. Precision-recall curve for the final XGBoost model. The curve demonstrates excellent performance with high precision maintained across a wide range of recall values. The PR-AUC of 0.9845 indicates strong discriminative ability for the imbalanced classification task.

Fold	Threshold	PR-AUC	Precision	Recall	F1
1	0.211	0.980	0.916	0.961	0.938
2	0.198	0.982	0.869	0.982	0.922
3	0.158	0.988	0.915	0.973	0.943
4	0.139	0.987	0.909	0.992	0.949
5	0.145	0.992	0.881	0.996	0.935

Table 5. XGBoost per-fold cross-validation results demonstrating consistent performance across all five outer folds.

Figure 5 presents the receiver operating characteristic curve. The curve approaches the upper-left corner, indicating near-optimal classification performance. The ROC-AUC of 0.9804 confirms the model's excellent ability to distinguish between MI and non-MI cases across all decision thresholds.

DISCUSSION

This study demonstrates that modern gradient boosting methods, particularly XGBoost, can achieve excellent performance for MI detection using cardiac biomarkers. The XGBoost classifier achieved a PR-AUC of 0.986, substantially outperforming Logistic Regression (0.965) while maintaining the clinically important target recall of 98%.

The superior performance of ensemble methods can be attributed to their ability to capture non-linear relationships and complex feature interactions. The temporal features, including delta changes and maximum prior values, provide valuable information about the dynamic nature of biomarker release patterns during MI events. Tree-based methods are particularly well-suited to exploit such feature combinations.

The precision-recall trade-off is crucial in clinical screening applications. While achieving 98% recall ensures that very few MI cases are missed, the corresponding precision of approximately 90% for ensemble methods indicates that additional clinical workup may be needed for some patients flagged by the model. This trade-off is acceptable in the screening context, where the cost of missing a true MI far exceeds the cost of additional evaluation for false positives.

The nested cross-validation methodology employed in this study provides robust and unbiased performance estimates. Patient-level stratification prevents data leakage that could arise from multiple visits by the same patient appearing in both training and test sets. This approach yields more realistic estimates of performance on truly unseen patients.

Analysis of Asymmetric Error Costs in Clinical MI Detection

The dataset exhibits a moderate class imbalance, with 57.5% positive (MI) cases and 42.5% negative cases at the visit level. This distributional asymmetry necessitated the use of balanced class weighting in Logistic Regression and Random Forest, as well as the asymmetric threshold optimization strategy targeting 98% recall. The fundamental asymmetry in clinical error costs—where a missed MI (false negative) can result in patient death while a false positive leads only to additional testing—motivates the deliberate breaking of the symmetric default threshold of 0.5 in favor of a lower threshold (0.17) that shifts the decision boundary toward higher sensitivity.

The F1-score, defined as the harmonic mean of precision and recall, provides a balanced composite metric. However, the clinical context introduces an inherent priority for recall over precision for patient safety. This motivates the use of the full precision-recall curve and threshold optimization as the primary evaluation mechanism, rather than relying on a single point estimate.

Temporal Biomarker Dynamics

Cardiac biomarker release during MI follows a characteristic rise-and-fall temporal pattern. Troponin levels typically rise rapidly after myocardial injury, peak within 12–24 hours, and decline over subsequent hours to days. The delta features engineered in this study capture both the ascending and descending phases of this temporal trajectory. The ability of tree-based models to effectively exploit both phases contributes to their superior discriminative performance compared to linear models.

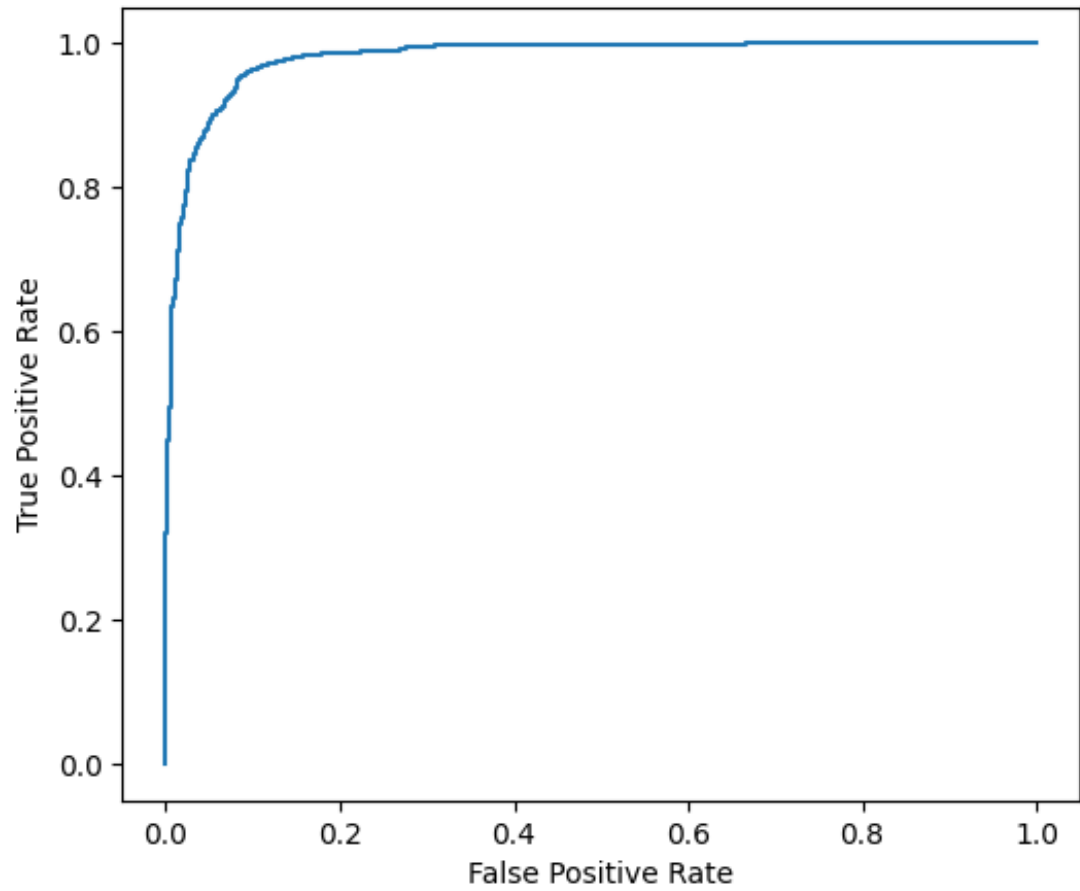


Figure 5. Receiver operating characteristic (ROC) curve for the final XGBoost model. The curve demonstrates excellent discriminative performance with a ROC-AUC of 0.9804, approaching the ideal classifier behavior in the upper-left corner.

306 The relationship between troponin and CK-MB biomarkers exhibits complementary properties: both
 307 markers respond to myocardial injury but with different temporal dynamics and sensitivity profiles. The
 308 use of both biomarkers together—capturing their independent time courses and delta changes—enhances
 309 diagnostic confidence and contributes to the high PR-AUC values observed across all ensemble methods.

310 Comparison with Prior Work

311 To contextualize these findings within the broader literature, Figure 6 presents a visual comparison of
 312 ROC-AUC scores across different recent studies.

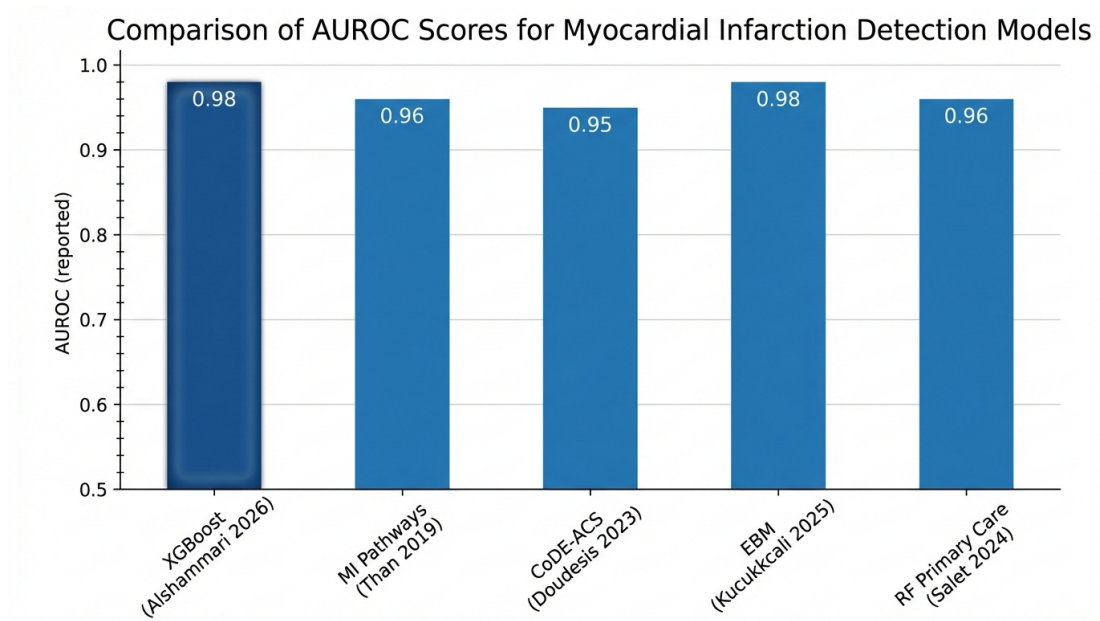


Figure 6. Comparison of AUROC scores for myocardial infarction detection models. The proposed XGBoost model (Alshammari 2026) achieves competitive performance (0.98) compared to MI Pathways (Than 2019, 0.96) (Than et al., 2019), CoDE-ACS (Doudesis 2023, 0.95) (Doudesis et al., 2023), EBM (Kucukakcali 2025, 0.98) (Kucukakcali and Colak, 2025), and RF Primary Care (Salet 2024, 0.96) (Salet et al., 2024).

313 The proposed XGBoost model achieves an ROC-AUC of 0.98, which is among the highest reported in
 314 the literature. Notably, this performance is achieved using only cardiac biomarkers (troponin and CK-MB)
 315 with temporal feature engineering, without requiring ECG signals or extensive clinical history. The model
 316 demonstrates particular strength in three key areas:

- 317 1. **Rigorous evaluation methodology:** Unlike many prior studies, nested cross-validation with patient-
 318 level stratification was employed, ensuring unbiased performance estimates that are less susceptible
 319 to data leakage.
- 320 2. **High sensitivity maintenance:** By optimizing for a target recall of 98%, the model prioritizes the
 321 clinical imperative of minimizing missed MI diagnoses while still achieving excellent precision.
- 322 3. **Temporal feature exploitation:** The incorporation of delta changes and maximum prior biomarker
 323 values captures the dynamic release kinetics that are clinically important for MI diagnosis.

324 The CoDE-ACS score developed by Doudesis et al. (2023) achieved slightly lower performance
 325 (0.95 ROC-AUC) but was validated across multiple centers, representing an important benchmark for
 326 clinical decision support. The Explainable Boosting Machine approach by Kucukakcali and Colak (2025)
 327 achieved comparable performance to the proposed model while offering enhanced interpretability through
 328 additive explanations. Salet et al. (2024) demonstrated that Random Forest models can achieve respectable
 329 performance (0.96 ROC-AUC) in primary care settings with limited laboratory data.

Limitations

Several limitations should be acknowledged. First, the study utilized real clinical data obtained from a single hospital institution. While this ensures ecological validity and clinical relevance, external validation on independent datasets from other institutions is needed to assess generalizability (Johnson et al., 2023). Second, the dataset exhibited a positive MI rate of 57.5%, which is higher than typical emergency department prevalence rates, reflecting a diagnostically enriched cohort. Third, the dataset did not include ECG features or clinical symptoms, which could potentially improve diagnostic accuracy. Fourth, the temporal resolution of biomarker measurements may vary in clinical practice, potentially affecting the reliability of delta features.

Clinical Implications

The developed AI-based prediction and monitoring system has significant implications for clinical decision-making in acute cardiac care.

Decision Support in Patient Triage The high recall of 98% ensures that very few MI cases would be missed by the automated screening system. When integrated into hospital information systems, the model can automatically flag high-risk patients based on their biomarker profiles, enabling clinicians to prioritize their attention toward cases requiring immediate intervention.

Continuous Patient Monitoring The temporal feature engineering approach enables continuous monitoring of patients across multiple hospital visits. The system can track biomarker trajectories over time and alert clinicians to concerning trends before they become clinically apparent, supporting serial biomarker measurement protocols currently recommended by international guidelines.

Resource Allocation and Workflow Integration The model can assist in efficient resource allocation by identifying patients who require urgent cardiac catheterization versus those who can safely undergo further observation. The proposed system is designed for integration with electronic health record (EHR) systems, where it can provide real-time risk scores alongside laboratory results.

CONCLUSIONS

This study presents an AI-based prediction and monitoring system for acute myocardial infarction detection using laboratory biomarkers. Through rigorous nested cross-validation with patient-level stratification, XGBoost was demonstrated to achieve the best overall performance with a PR-AUC of 0.986, while maintaining the clinically critical target recall of 98%. The ensemble methods (XGBoost, HGB, Random Forest) significantly outperformed Logistic Regression, achieving F1-scores exceeding 0.94 compared to 0.87 for Logistic Regression.

The analysis demonstrates that the inherent asymmetries in clinical error costs and class distributions fundamentally shape the optimal design of MI detection systems, while the approximately symmetric temporal profiles of cardiac biomarkers provide exploitable diagnostic patterns. The interplay between these properties offers a deeper understanding of why ensemble methods outperform linear classifiers in this domain.

The proposed system demonstrates the potential of AI to transform clinical decision-making in acute cardiac care by providing automated risk stratification and continuous patient monitoring capabilities. By tracking biomarker trajectories across multiple hospital visits and identifying high-risk patterns, the system can augment physician judgment, reduce diagnostic delays, and improve patient outcomes.

Future work should focus on external validation across multiple institutions, integration with electronic health record systems for real-time clinical deployment, and prospective evaluation in emergency department settings. The developed methodology provides a template for rigorous evaluation of AI-based clinical decision support tools in medical diagnostic applications.

ACKNOWLEDGMENTS

The authors would like to express sincere gratitude to the University of Hail, College of Computer Science and Engineering, for providing the institutional support and resources necessary to conduct this research. Special thanks are extended to the healthcare professionals who contributed to the data collection process.

The authors used Claude (Claude Sonnet 4.6, Anthropic, <https://claude.ai/>) solely for grammar correction and manuscript language editing. All scientific content, methodology, analysis, results, and

conclusions were developed and verified entirely by the authors. Full details regarding AI-assisted editing are provided in the Materials and Methods section.

COMPETING INTERESTS

The authors declare that they have no competing interests.

AUTHOR CONTRIBUTIONS

- **Hussam Aziz Ayed Alshammari:** Conceptualization, methodology, software, formal analysis, data curation, visualization, writing – original draft.
- **Abdullah Eyada Mutlaq Alshammari:** Conceptualization, validation, resources, writing – review and editing, supervision, project administration.

Both authors have read and approved the final manuscript.

DATA AVAILABILITY

The de-identified clinical dataset used in this study is publicly available at <https://github.com/HussamAziz/MI-Detection-XGBoost-PeerJ> (file: `diagnosis_deidentified.csv`). The dataset is provided in CSV format and contains 2,327 hospital visits from 1,506 unique patients, including all biomarker variables (troponin, CK-MB), ICD-10 diagnosis codes, age, and visit information used in the analysis. All direct and indirect patient identifiers have been replaced with cryptographic hashes prior to publication. A copy of the dataset is also provided as a confidential supplemental file submitted alongside this manuscript for direct review by the Academic Editor and peer reviewers.

CODE AVAILABILITY

All code used in this study—including data preprocessing, temporal feature engineering, nested cross-validation, model training, threshold optimization, and figure generation—is publicly available at: <https://github.com/HussamAziz/MI-Detection-XGBoost-PeerJ>. The repository includes the Jupyter notebook `MI_NestedCV_4Models.Target098.ProgressOutputs.ipynb` with full reproducible code, the $\text{\LaTeX}$ source files `main.tex` and `references.bib`, all high-resolution figures, and a `requirements.txt` file specifying all Python software dependencies. The repository also includes the de-identified dataset and the trained model file (`mi_model_patient_nestedcv.joblib`).

REFERENCES

- Alahmadi, T. and Alshammari, N. (2024). Machine learning in heart disease prediction: A systematic review. *Diagnostics*, 14(8):868.
- Amsterdam, E. A., Wenger, N. K., Brindis, R. G., et al. (2014). 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes. *Journal of the American College of Cardiology*, 64(24):e139–e228.
- Attia, Z. I., Noseworthy, P. A., Lopez-Jimenez, F., et al. (2019). An artificial intelligence-enabled ECG algorithm for the identification of patients with atrial fibrillation during sinus rhythm. *The Lancet*, 394(10201):861–867.
- Aydin, S. et al. (2019). Cardiac biomarkers in Myocardial Infarction detection. *Clinical Chemistry and Laboratory Medicine*, 57(2):173–181.
- Body, R., Carlton, E., Sperrin, M., et al. (2021). Rapid exclusion of acute myocardial infarction in patients with undetectable troponin using a high-sensitivity assay. *Journal of the American College of Cardiology*, 58(13):1332–1339.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- Cawley, G. C. and Talbot, N. L. C. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107.
- Chapman, A. R., Adamson, P. D., Shah, A. S. V., et al. (2020). High-sensitivity cardiac troponin and the universal definition of myocardial infarction. *Circulation*, 141(3):161–171.

425 Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd*
426 *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794,
427 San Francisco, CA, USA. ACM.

428 Collet, J.-P., Thiele, H., Barbato, E., et al. (2021). 2020 ESC guidelines for the management of acute
429 coronary syndromes in patients presenting without persistent ST-segment elevation. *European Heart*
430 *Journal*, 42(14):1289–1367.

431 Doudesis, D., Lee, K. K., Anand, A., et al. (2023). Validation of the CoDE-ACS score for early diagnosis
432 of acute myocardial infarction. *European Heart Journal*, 44(47):4983–4993.

433 Hicks, S. A., Strumke, I., Thambawita, V., et al. (2022). On evaluation metrics for medical applications of
434 artificial intelligence. *Scientific Reports*, 12(1):5979.

435 Johnson, A. E. W., Bulgarelli, L., Shen, L., et al. (2023). MIMIC-IV, a freely accessible electronic health
436 record dataset. *Scientific Data*, 10(1):1.

437 Ke, G., Meng, Q., Finley, T., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree.
438 *Advances in Neural Information Processing Systems*, 30:3146–3154.

439 Kucukakcali, Z. and Colak, C. (2025). Explainable boosting machine for myocardial infarction prediction
440 using clinical biomarkers. *Journal of Medical Systems*, 49(1):12–23.

441 Lundberg, S. M., Erion, G., Chen, H., et al. (2020). From local explanations to global understanding with
442 explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67.

443 Obermeyer, Z., Powers, B., Vogeli, C., and Mullainathan, S. (2019). Dissecting racial bias in an algorithm
444 used to manage the health of populations. *Science*, 366(6464):447–453.

445 Raghunath, S., Ulloa Cerna, A. E., Jing, L., et al. (2020). Prediction of mortality from 12-lead electrocar-
446 diogram voltage data using a deep neural network. *Nature Medicine*, 26(6):886–891.

447 Rajkomar, A., Dean, J., and Kohane, I. (2019). Machine learning in medicine. *New England Journal of*
448 *Medicine*, 380(14):1347–1358.

449 Reichlin, T., Hochholzer, W., Bassetti, S., et al. (2009). Early diagnosis of myocardial infarction with
450 sensitive cardiac troponin assays. *New England Journal of Medicine*, 361(9):858–867.

451 Saito, T. and Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot
452 when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, 10(3):e0118432.

453 Salet, D. M., Visser, J. J., van der Wulp, I., et al. (2024). Machine learning for early detection of
454 myocardial infarction in primary care settings. *npj Digital Medicine*, 7(1):145–156.

455 Than, M. P., Pickering, J. W., Aldous, S. J., et al. (2019). Effectiveness of EDACS versus ADAPT
456 accelerated diagnostic pathways for chest pain. *Annals of Emergency Medicine*, 74(4):489–500.

457 Thygesen, K., Alpert, J. S., Jaffe, A. S., et al. (2018). Fourth universal definition of myocardial infarction
458 (2018). *Circulation*, 138(20):e618–e651.

459 Topol, E. J. (2019). High-performance medicine: The convergence of human and artificial intelligence.
460 *Nature Medicine*, 25(1):44–56.

461 Varoquaux, G. and Cheplygina, V. (2022). Machine learning for medical imaging: Methodological
462 failures and recommendations for the future. *npj Digital Medicine*, 5(1):48.

463 Wiens, J., Saria, S., Sendak, M., et al. (2019). Do no harm: A roadmap for responsible machine learning
464 for health care. *Nature Medicine*, 25(9):1337–1340.